Mohamad Chehade

Research Curriculum Vitae

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Research focus: trustworthy AI for consequential decisions. I develop methods to verify safety, adapt behavior at inference time, and evaluate models by downstream decision quality. First-author papers at NeurIPS and ICML; research at Argonne and Los Alamos National Laboratories.

Education

The University of Texas at Austin (Austin, TX)
Ph.D., Electrical and Computer Engineering | Aug. 2023 – Present
Master of Science in Engineering, Aug. 2023 – May 2026. Advisor: Dr. Hao Zhu.

American University of Beirut (Beirut, Lebanon)
B.Eng., Electrical and Computer Engineering | Aug. 2019 – Jun. 2023
GPA: 4.0/4.0; Minor in Mathematics. Final-year project: Optimal Power Flow via Machine Learning (Dr. Rabih Jabr).
Undergraduate research: Microgrid Sizing using Ordinal Optimization (Dr. Sami Karaki).

Skills and Relevant Coursework

Programming: Python, C/C++, MATLAB, SQL, R
ML and scientific computing: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Matplotlib
Optimization: Gurobi; convex and mixed-integer optimization, ordinal optimization
Power systems: PowerWorld, MATPOWER, Simulink, SPICE, HOMER, PVSyst, LabVIEW
Development: Git, Linux, LaTeX, Overleaf, Microsoft Office Suite
Languages: English (fluent), French (intermediate), Arabic (native)
Coursework: Reinforcement Learning; Generative AI; Statistical Machine Learning; Applied Machine Learning; Probability and Stochastic Processes; Applied Stochastic Processes; Learning-based Optimal Control; Convex Optimization; Energy Optimization and Operation

Research and Engineering Experience

The University of Texas at Austin (Austin, TX)
Graduate Research Assistant – Advisor: Dr. Hao Zhu | Aug. 2023 – Present

- **Safe AI agents (TRAM, NeurIPS 2026):** Let already-trained reinforcement-learning agents follow safety rules that are only specified at deployment. The method scores each pretrained agent on the new reward and the new risk, then at each step takes the action of the agent with the best risk-adjusted score, with no retraining. It lowered risk while keeping reward in robotics simulators (MuJoCo, Safety-Gymnasium) and an LLM alignment task.
- **Safer LLM responses (SITAlign, ICML 2025):** Made LLM answers as helpful as possible while keeping them harmless enough. Instead of retraining, the method steers the model's text generation at inference time to maximize the main goal (helpfulness) subject to a minimum threshold on the others (harmlessness). It beat the best multi-objective decoding method by 22.3% in GPT-4 win-tie rate on helpfulness.
- **LLMs for grid emergencies (PESGM 2026):** Taught an LLM to choose which power lines to switch off during wildfire shutoffs to limit outages. The model first learns to imitate an optimization solver (supervised fine-tuning), is then refined with preferences ranked by a full power-flow check (DPO), and finally generates several plans and keeps the best feasible one (best-of-N). This cut power-flow failures from 50% to single digits on a 118-bus grid.
- **Grid voltage control (NEO-Grid, HICSS 2026):** Kept distribution-grid voltages within limits as solar and batteries fluctuate. The method trains neural networks to predict voltages from power injections and uses them inside the optimization in place of the usual linear approximation; for real-time control, it models the feedback loop between voltages and inverters as a deep equilibrium model solved directly for its steady state. It regulated voltage better than standard linear and heuristic baselines.
- **Battery control (NAPS 2024):** Studied how to charge and discharge a battery as electricity prices change, comparing a reinforcement-learning controller that learns from data with a model-based controller that plans using a battery model, to show when each approach is the better choice.
- **Carbon-aware energy management (PESGM 2026):** Co-developed a building energy manager that weighs electricity cost against carbon emissions. It schedules grid purchases, battery storage, and flexible demand with a mixed-integer program that includes the grid's hour-by-hour carbon intensity, re-planned in real time from a Transformer forecast of prices and carbon. On real PJM market data it cut emissions by 22.5% for a 1.7% cost increase.

Argonne National Laboratory (Remote)
Research Aide – Grid Optimization and Infrastructure Group | May – Aug. 2026
Mentors: Dr. Feng Qiu and Dr. Wei Gao.

- Adapted pretrained time-series foundation models so their forecasts lead to cheaper decisions, not just smaller forecast errors. The pretrained model stays frozen and a small add-on module is trained on the cost of the decisions made with its forecasts in downstream optimization problems.

Los Alamos National Laboratory (Remote)
Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics | May – Aug. 2025
Mentor: Dr. Russell Bent.

- Developed training methods that use the input regions where a network is proven to behave correctly (from LEVIS) to make neural networks provably more robust to input perturbations.

Los Alamos National Laboratory (Los Alamos, NM)
Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics | Jun. – Aug. 2024
Mentors: Dr. Wenting Li and Dr. Brian Bell.

- Developed two algorithms (LEVIS, ICML 2025) that find large input regions where a neural network is guaranteed to behave correctly. The first uses mixed-integer optimization to find the closest input where the network's answer flips and builds the largest safe ball up to it; the second covers the whole safe region with many such balls. A faster reformulation gave up to 6x speedup on power-flow and image-classification models.

University of Connecticut – PEARL Lab (Storrs, CT)
Research Intern – Center for Clean Energy Engineering | Jun. – Aug. 2022
Advisor: Dr. Ali Bazzi.

- Developed inverter fault diagnosis using combinational logic with industrial motor manufacturer Ward Leonard.
- Optimized and built an inverter circuit for high-power testing using mixed-integer linear programming.

Publications – Published or Accepted

1. **M. Chehade**, A. S. Bedi, S. Chakraborty, A. Zhang, and H. Zhu. Test-Time Risk Adaptation with Mixture of Agents. *NeurIPS 2026 (accepted)*.
2. **M. Chehade**, W. Li, B. W. Bell, R. Bent, S. R. Kazi, and H. Zhu. LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *ICML 2025*.
3. **M. Chehade**, S. S. Ghosal, S. Chakraborty, A. Reddy, D. Manocha, H. Zhu, and A. S. Bedi. Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *ICML 2025*.
4. **M. Chehade** and H. Zhu. Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. *IEEE PES General Meeting (PESGM), Montréal, Canada, 2026*.
5. **M. Chehade** and H. Zhu. NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Hawaii International Conference on System Sciences (HICSS), 2026*.
6. Y.-H. Cho, **M. Chehade**, F. Al-Janahi, S. Lim, J. Mohammadi, and H. Zhu. Carbon-aware Market Participation for Building Energy Management Systems. *IEEE PES General Meeting (PESGM), Montréal, Canada, 2026*.
7. **M. Chehade** and S. Karaki. BOOST: Microgrid Sizing Using Ordinal Optimization. *IEEE Texas Power and Energy Conference (TPEC), 2025, pp. 1–4*.
8. **M. Chehade**, Y.-H. Cho, S. Chinchali, and H. Zhu. Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *North American Power Symposium (NAPS), 2024, pp. 1–5*.

Awards and Honors

2026 Third place, American Society of Mechanical Engineers (ASME) CIE Student Hackathon – AI for engineering design and manufacturing.
2024 Best Graduate Presentation Award, North American Power Symposium (NAPS).
2023 Mohamad Ali Safieddine Award for Academic Excellence – ranked first across the American University of Beirut Faculty of Engineering and Architecture.
2023–present Cockrell School of Engineering Fellowship, UT Austin.
2023 Electrical and Computer Engineering Distinguished Graduate Award – ranked first among ECE graduates.
2023 Exceptional ECE Final Year Project Award – Power and Energy Systems.
2021 Best Pitch Award, EPFL Tech4Impact Summer School (Swiss Federal Institute of Technology Lausanne).
2019–2023 Dean's Honor List, American University of Beirut – every awarded semester.

Tutorial

October 26, 2026 – IEEE SmartGridComm, College Station, TX. Agentic AI for Grid Operators in Active Distribution Networks within a Multi-Agent Framework. Co-presenters: Dr. Alaa Selim (MIT) and Dr. Mahmoud Abdelaal (ABB).

Peer Review Service

Neural Information Processing Systems (NeurIPS): June 2026 – Present
International Conference on Machine Learning (ICML): Feb. 2026 – Present
International Conference on Learning Representations (ICLR): Oct. 2025 – Present
AAAI Conference on Artificial Intelligence: Aug. 2026 – Present
IEEE Transactions on Smart Grid: Sep. 2023 – Present
IEEE Power Engineering Letters: May 2026 – Present
IEEE Systems Journal: Sep. 2025 – Present
Hawaii International Conference on System Sciences (HICSS): June 2025 – Present
Asilomar Conference on Signals, Systems, and Computers: July 2024 – Present
IEEE Texas Power and Energy Conference (TPEC): Dec. 2024 – Present
Transactions on Machine Learning Research (TMLR): Oct. 2026 – Present